

We will start promptly at 1:37 and end at 2:17

Come to the front and grab eight sheets (DON'T PEAK)

You will turn only ONE sheet (the last one). Please have your SUID on the sheet you will turn in

All other sheets will not be submitted and simply specify the questions

There are eight questions

NO CHEAT SHEETS (you can not look at your reports)

Feel free to grab a pen and hi-chew

Allowed to take the quiz if you're waitlist or waitlistwaitlist or auditing

Quiz 1 Answers

Reminders

It is okay and expected to not get 100% on these quizzes. It is okay if this was uncomfortable, you'll have six more attempts to become familiar, and instructors are open to feedback about how to structure assignments and quizzes

Some calibration:

Instructors without running assignment-specific experiments: ~70%

Opus without running assignment-specific experiments: ~70%

Opus with running experiments and access to report: ~100%

Structure

For each question, we'll go over the following

- The ground-truth answer with an experiment that reveals it (examples of experiments you might want to run on your assignment)
- Some papers on these topics
 - The papers may or may not be accurate
 - We don't expect you have seen these papers. They're just easy experiments for us to point to
- Potential mental model updates
 - These updates will likely be flawed. It is a useful exercise for you to come up with examples where they are flawed.
 - Now that you have the foundations, a lot of this class will be testing existing mental models and developing your own!

Discussion

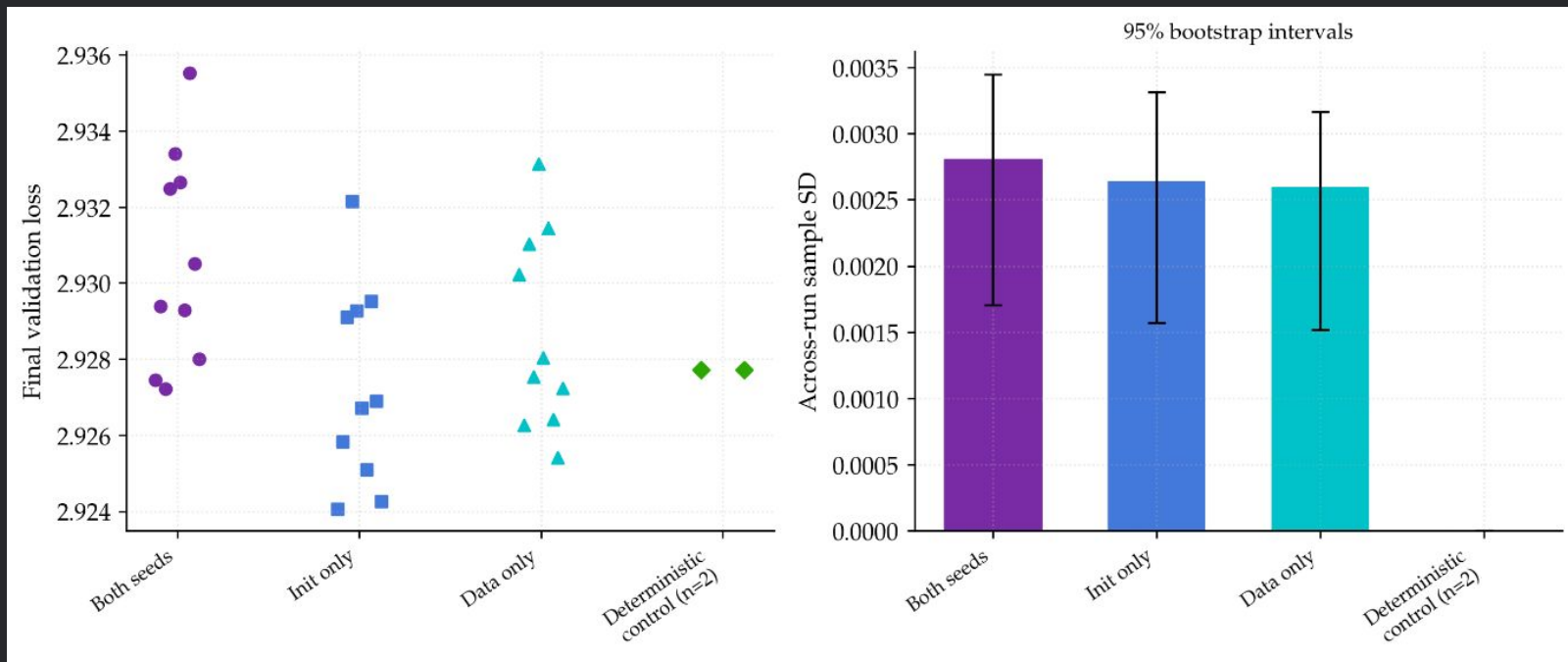
Please treat this as interactive! (subject to time constraints of explanation)

You as a class will have collectively run more experiments than the staff (maybe not yet)

Please send any feedback you have (on quiz/assignment/course design or anything else). Make sure to meet us if you feel that it'd be helpful for your learning

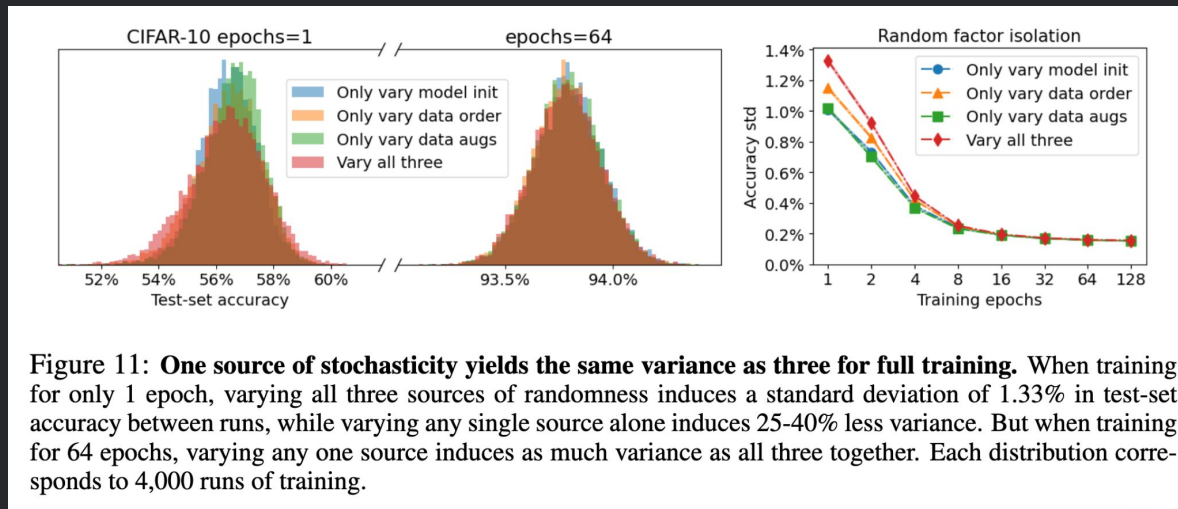
Question 1: What normal training noise looks like

Answer 1a) run-to-run std for varying both seeds was measured to be (0.002807)



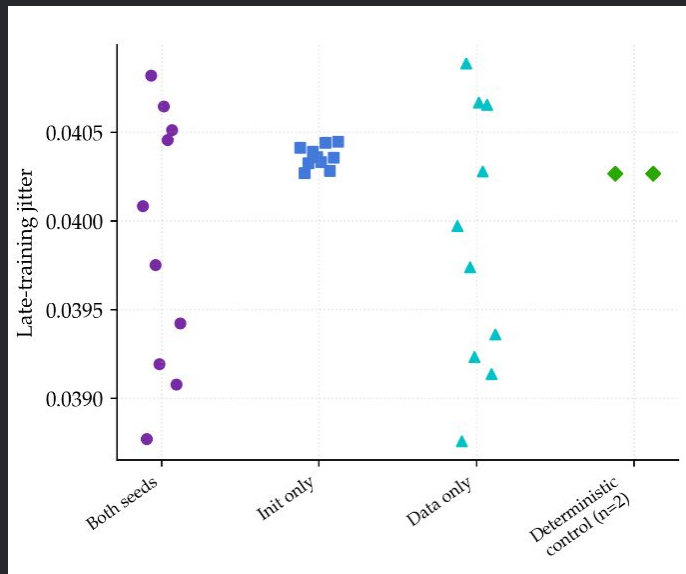
Question 1: What normal training noise looks like

This run-to-run std is roughly the same if you vary the data seed, init seed, or both. You can find more thorough analysis from [Summers and Dineen 2021](#), [Jordan 2024](#) where they find similar results. They seem to have more discrepancy at low train epochs



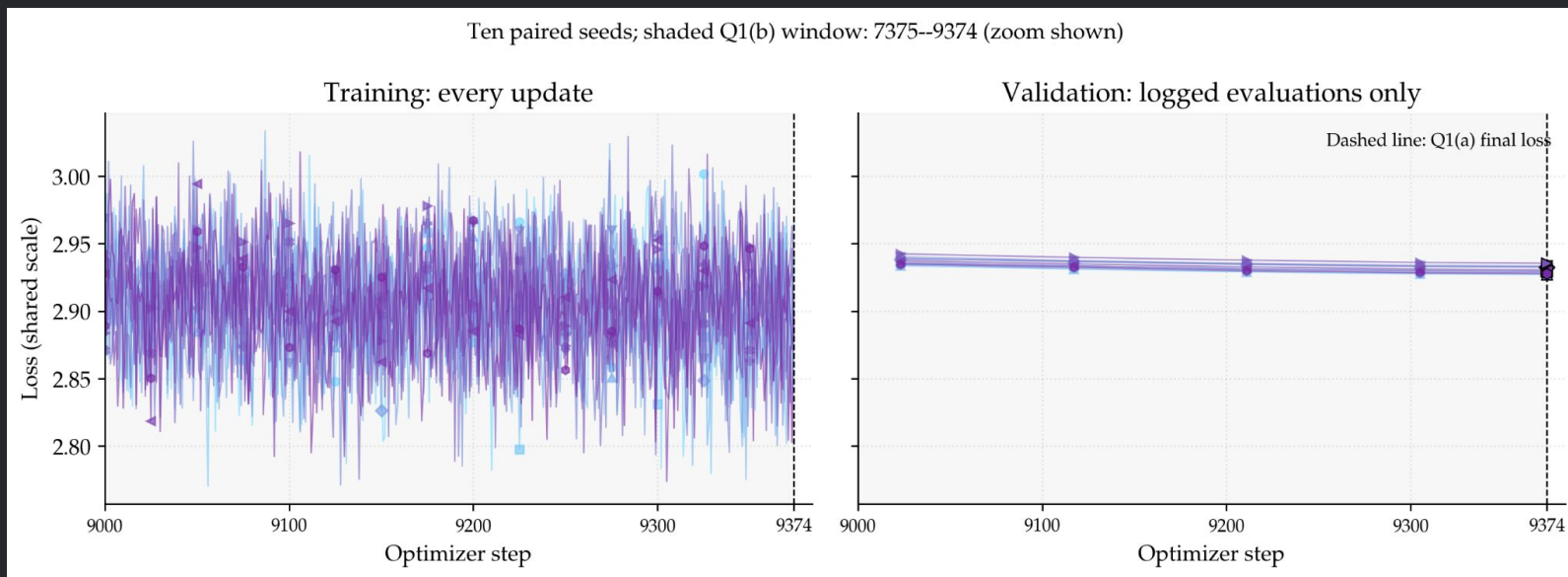
Question 1: What normal training noise looks like

Answer 1b) Jitter is close to 0.04. Plot below visualizes jitter for three sources of variation, finding it to be close to 0.04 in all settings



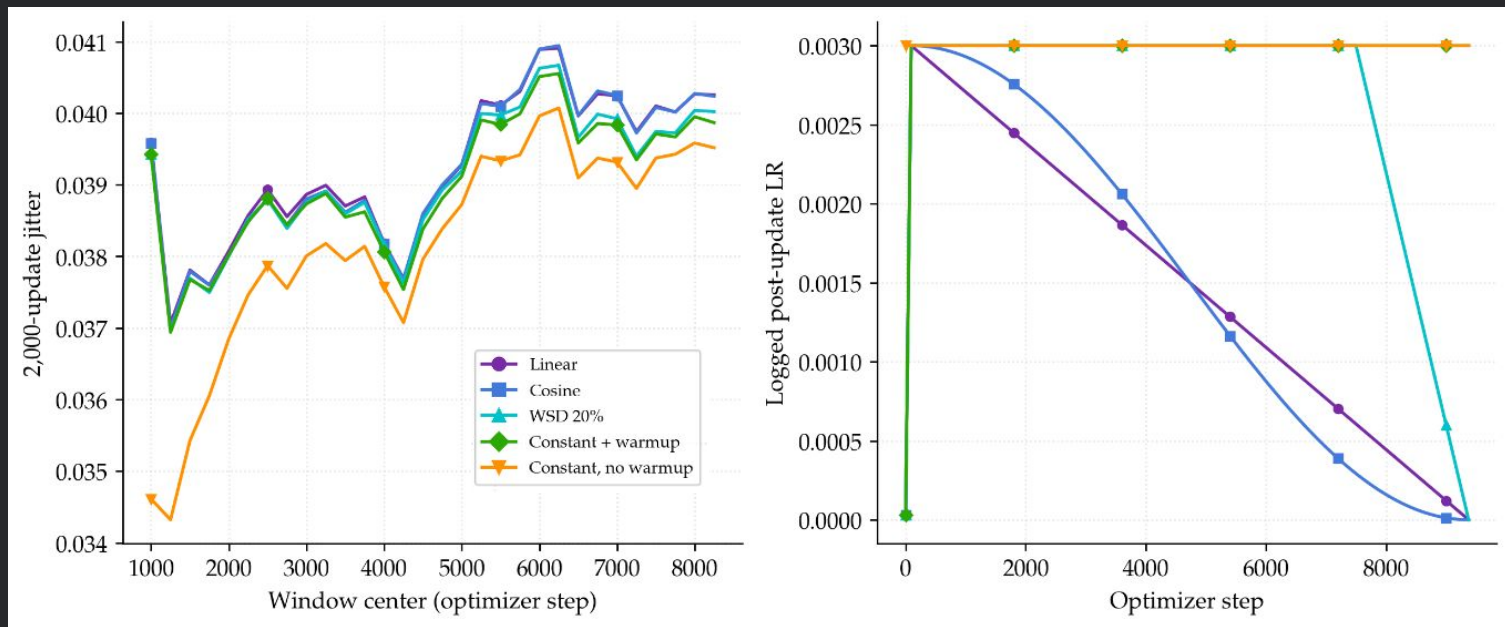
Question 1: What normal training noise looks like

Jitter for train loss vs test loss. This shows most of the variation in train jitter comes from the fact that samples keep changing



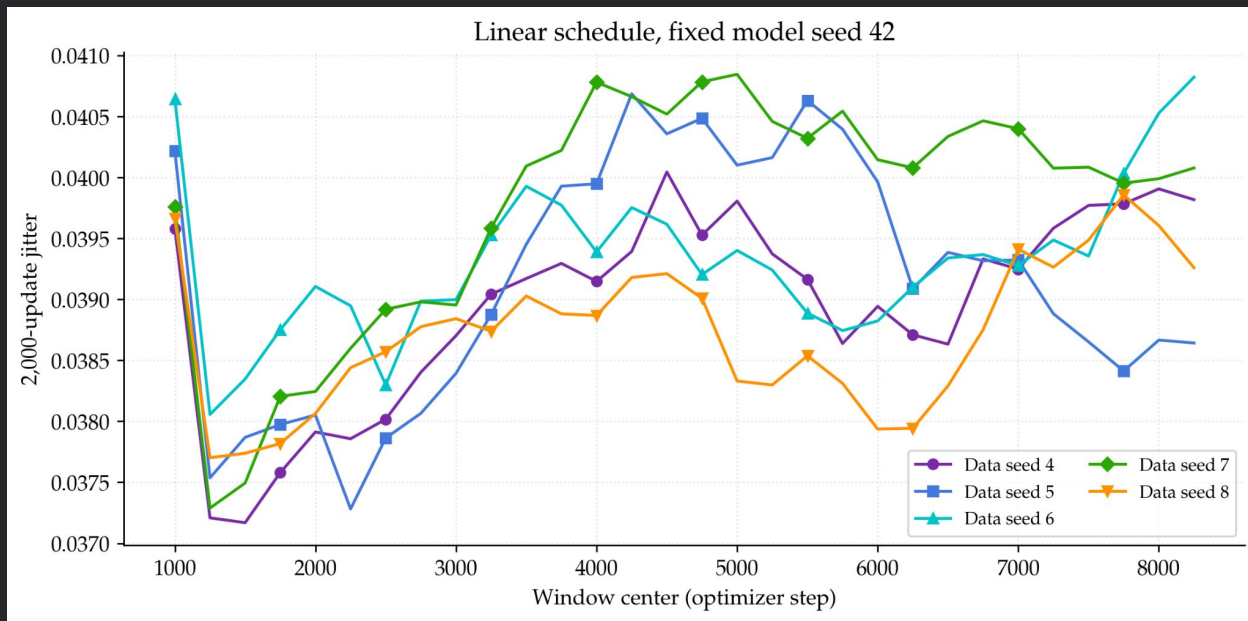
Question 1: What normal training noise looks like

Jitter might slightly increase over training? Learning rate schedule seems to have minimal role, so maybe it's an artifact of data order.



Question 1: What normal training noise looks like

When we try different data seeds, we still see similar shapes so it is a function of training dynamics



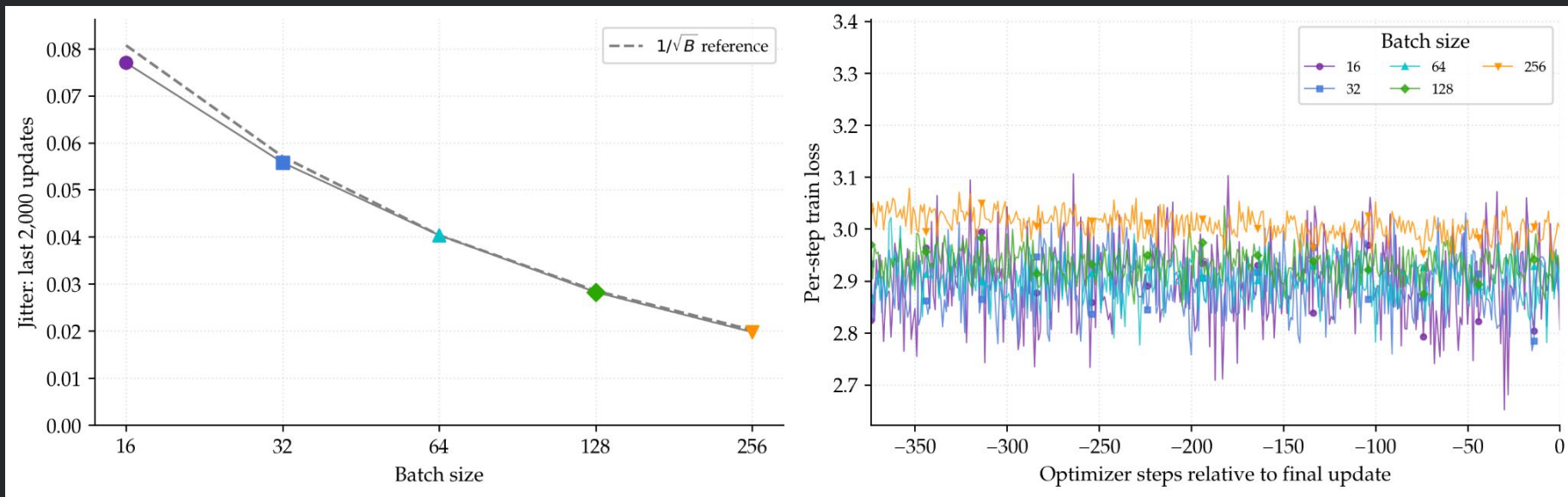
Question 1: What normal training noise looks like

Potential mental model updates:

1. Run-to-run variance is 0.003 in our setting and loss changes shouldn't be considered significant if they're around/below this
2. Neural network training is inherently unstable, and one source of variation is enough to expand the full range (more in question 8)
3. Measuring loss on the validation set is much less noisy than measuring loss on the train set.

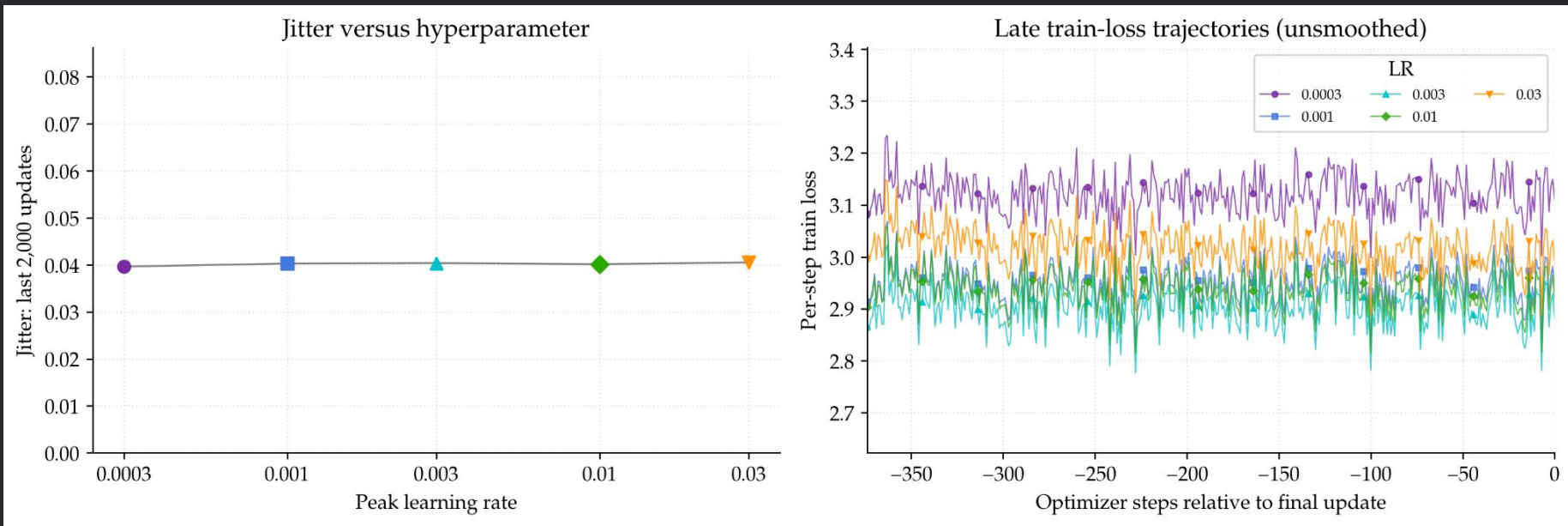
Question 2: Loss-curve microstructure

Answer 2) Batch size 16 has 2x jitter, Batch size 256 has 1/2x jitter



Question 2: Loss-curve microstructure

While batch size has a large impact on jitter, learning rate seems to have very little impact



Question 2: Loss-curve microstructure

We know that most of the training jitter is dominated by the fact that samples are different every batch. If you think about the examples in a batch being independent random variables, variance scales in the batch size and standard deviation scales in $\sqrt{\text{batch size}}$

Reasoning about the variance of your loss and gradient are really important for optimization analyses when setting hyperparameters. You will explore this in your next assignment on optimization hyperparameters! Extra reading: [Malladi 2024](#), [Zhang et al 2019](#)

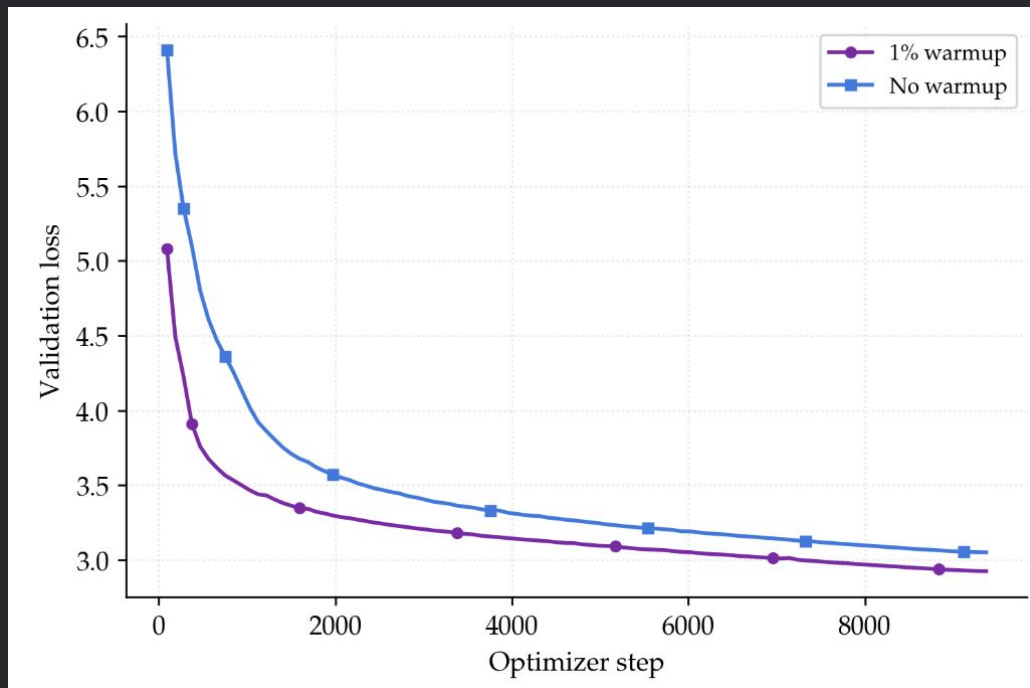
Question 2: Loss-curve microstructure

Potential mental model updates:

- Jitter comes from variance in the train data and is mostly controlled by batch size

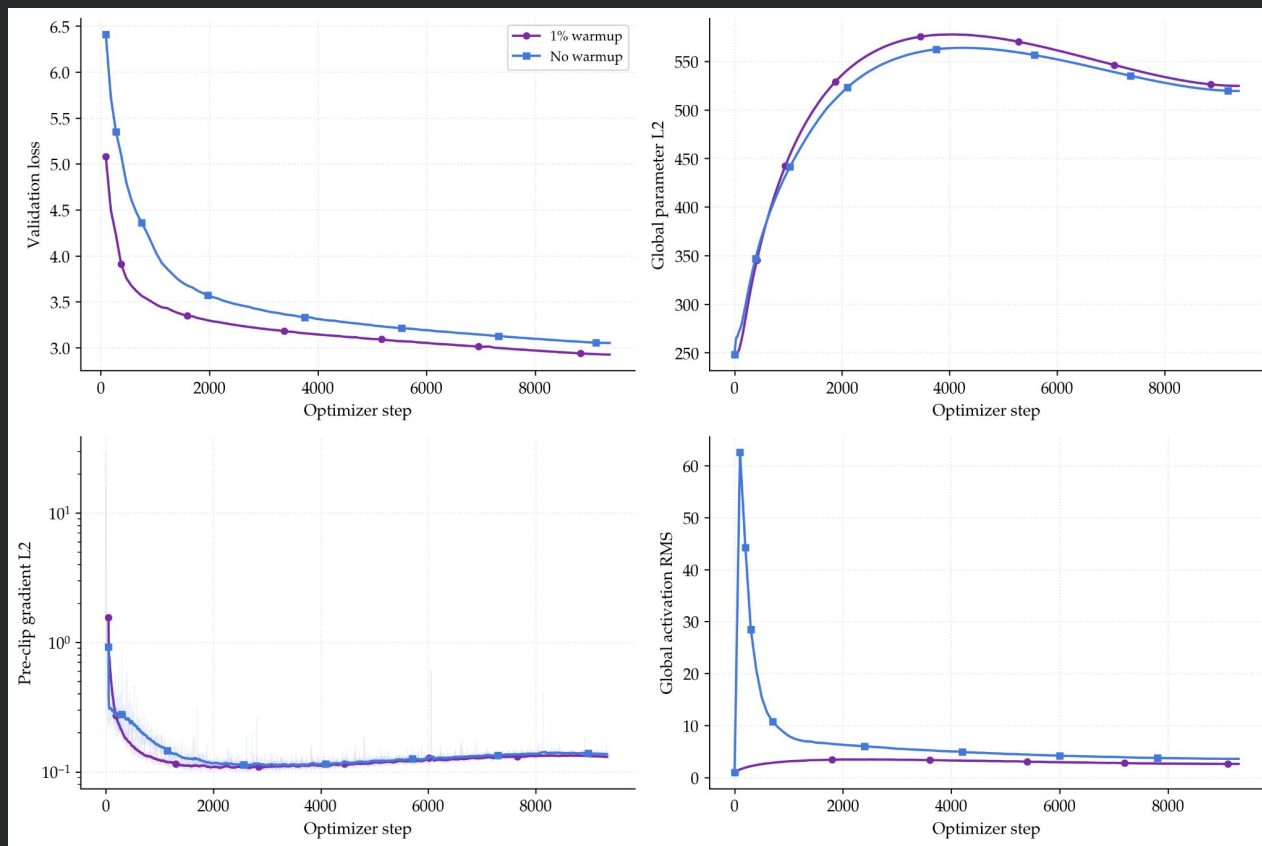
Question 3: Skipping warmup

Answer 3) Loss differential averages to 0.106 over 5 seeds, all 5 remain in 0.05-0.15 bucket



Question 3: Skipping warmup

Most of the parameter statistics look normal except for the activation RMS while the learning rate is still high



Question 3: Skipping warmup

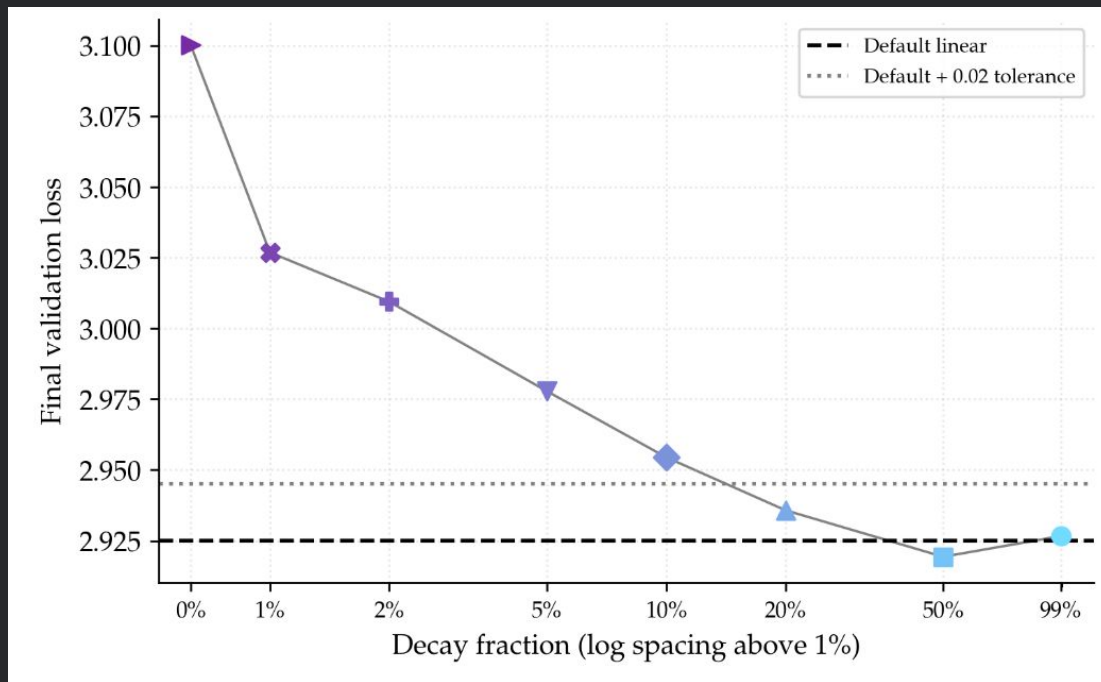
Potential mental model updates:

- We will cover more of these in a more thorough analysis in question 5

Related reading: [Wortsman et al, 2023](#)

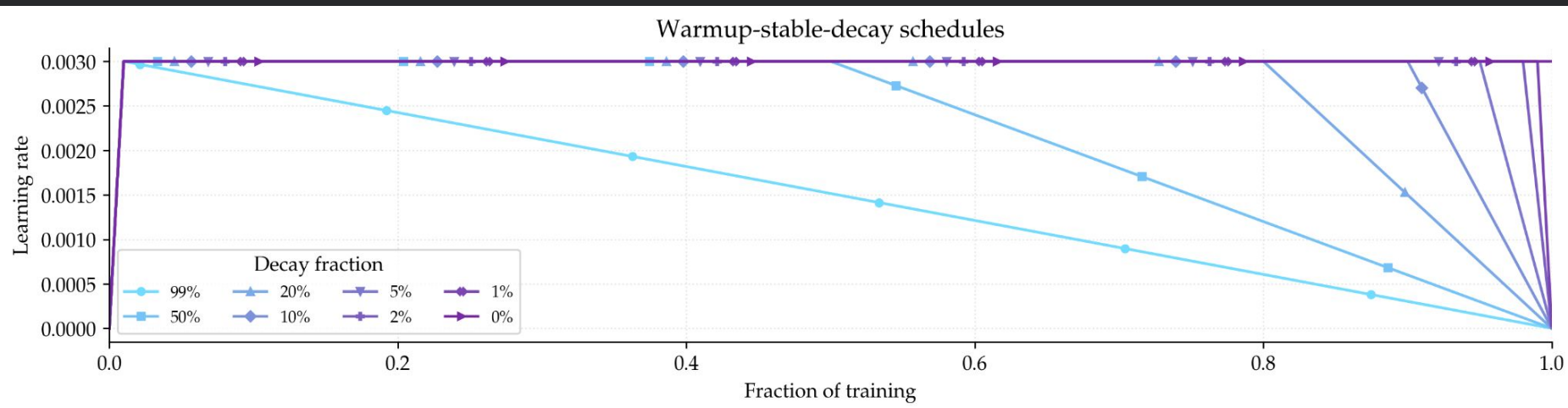
Question 4: Trimming the decay phase

Answer 4) the lowest cooldown period with 0.02 loss tolerance is 20%



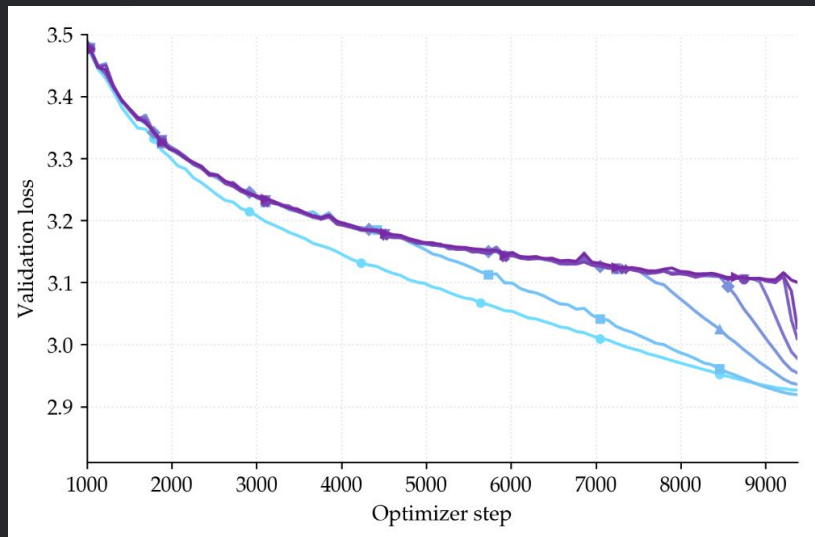
Question 4: Trimming the decay phase

Context: here are the decay schedules we asked about



Question 4: Trimming the decay phase

Here are the loss curves for these decay fractions. Notice there's a correspondence: slow progress for high lrs, rapid progress when lr decreases. However, too short of a decay period hurts loss

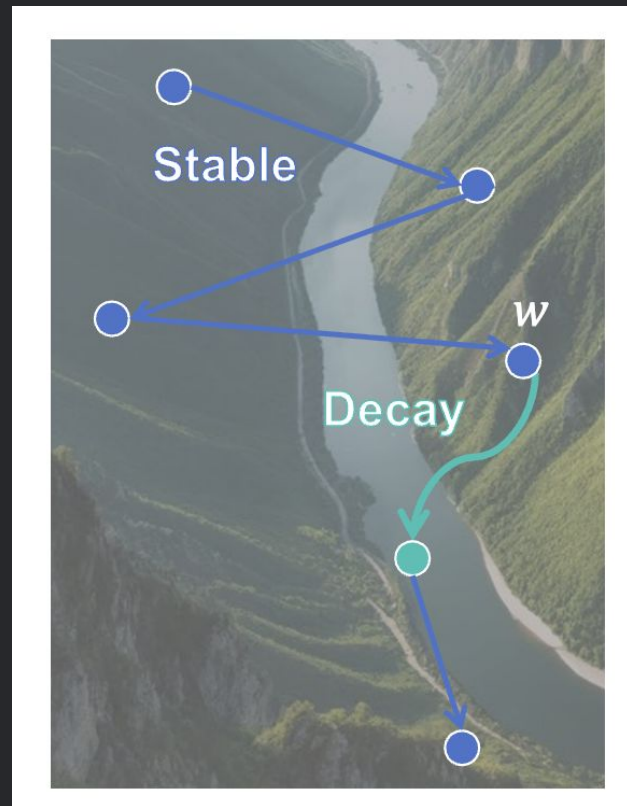


Question 4: Trimming the decay phase

There are multiple intuitions for why WSD schedules have this loss shape

[Wen et al, 2024](#): River-valley landscape. High learning rates bounce around the top of the valley while making progress along the river, low learning rates reach the bottom of the valley

[Schaipp et al, 2024](#): Predictions from convex optimization theory (next slide)



Question 4: Trimming the decay phase

Potential confounder: Previous works suggest WSD has different optimal learning rates from linear/cosine learning rate schedules according to convex optimization theory. Their experiments look like our plots!

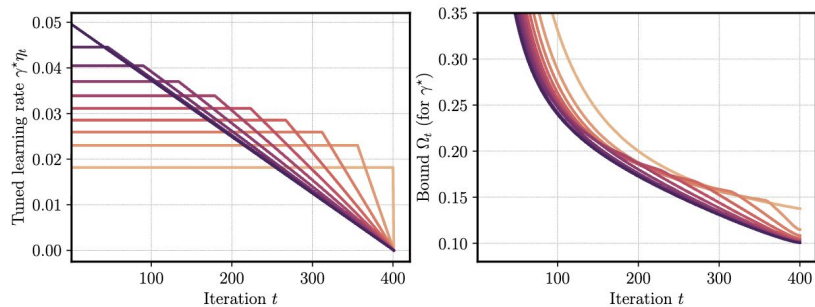
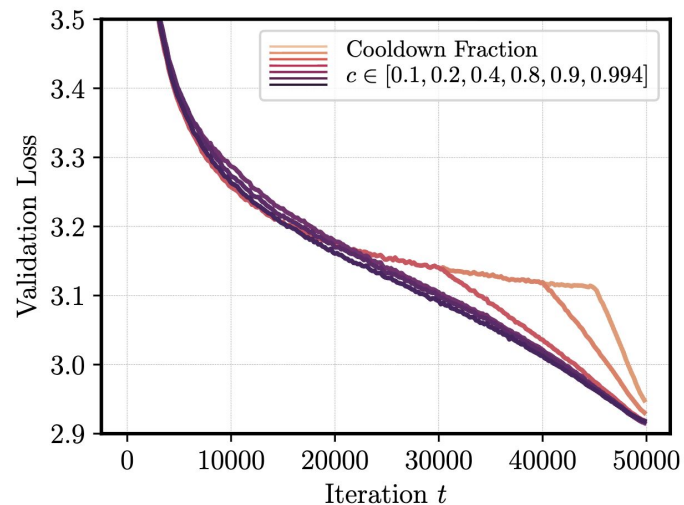


Figure 5. Schedule (**left**) and theoretical convergence (**right**) for varying cooldown fraction. With optimal base learning-rate γ^* , starting the cooldown at $T_0 = 1$ is optimal. [Fig. 21](#) shows the analogous plot for real experiments with the same behavior.



Question 4: Trimming the decay phase

Potential mental model updates

- WSD shape has a sharp annealing period
- Need some time to decay the schedule to get the benefit of WSD, for d8 this is close to 20% of training
 - Typical for some existing language models such as MiniCPM
 - (I had estimated around 10% of training would be sufficient when I was looking at the quiz)
- Larger variation in the loss early in the training trajectory relative to the end of the trajectory

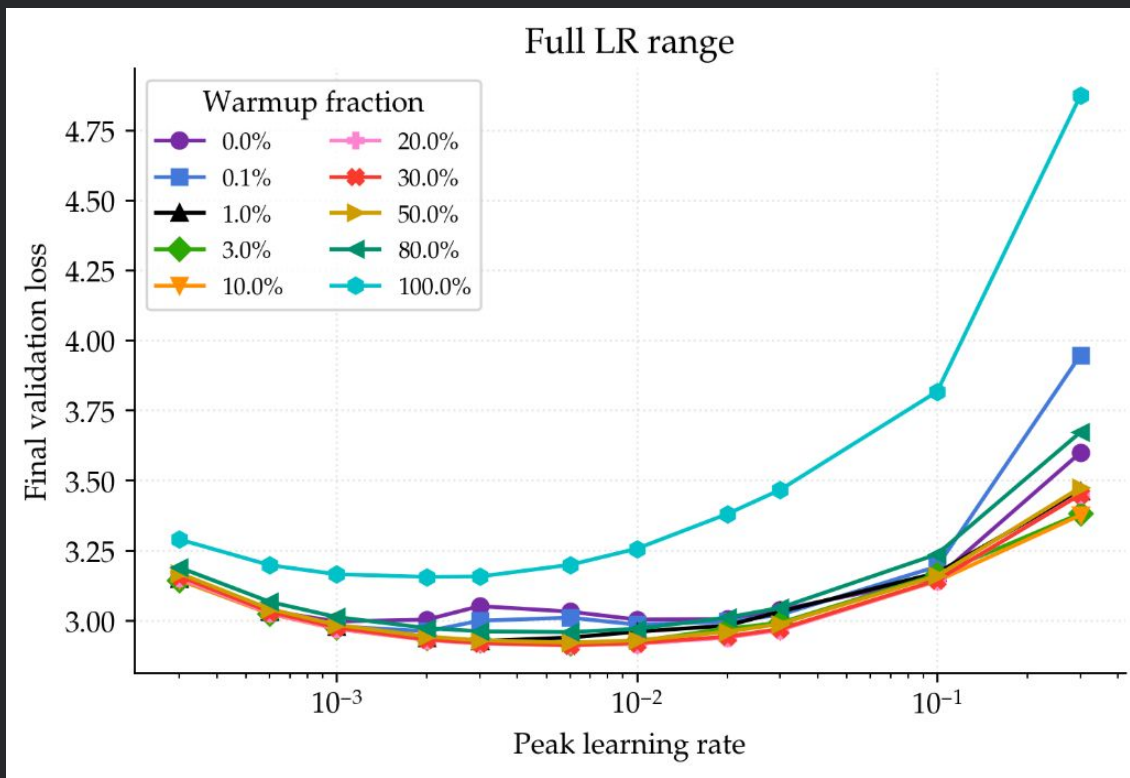
Question 5: Peak learning rate and warmup

Answer 5) LR 0.009 and 5% warmup is below 2.94 (in fact better than the baseline), LR 0.027 and 5/20% warmup is 2.94-3.05

LR	Warmup	Final loss	Answer
0.009	5%	2.91646	Below 2.94
0.027	5%	2.97501	2.94–3.05
0.027	20%	2.95563	2.94–3.05

Question 5: Peak learning rate and warmup

Here is loss as a function of warmup fraction and learning rate (for many more learning rates and warmup fractions than you as students can do)

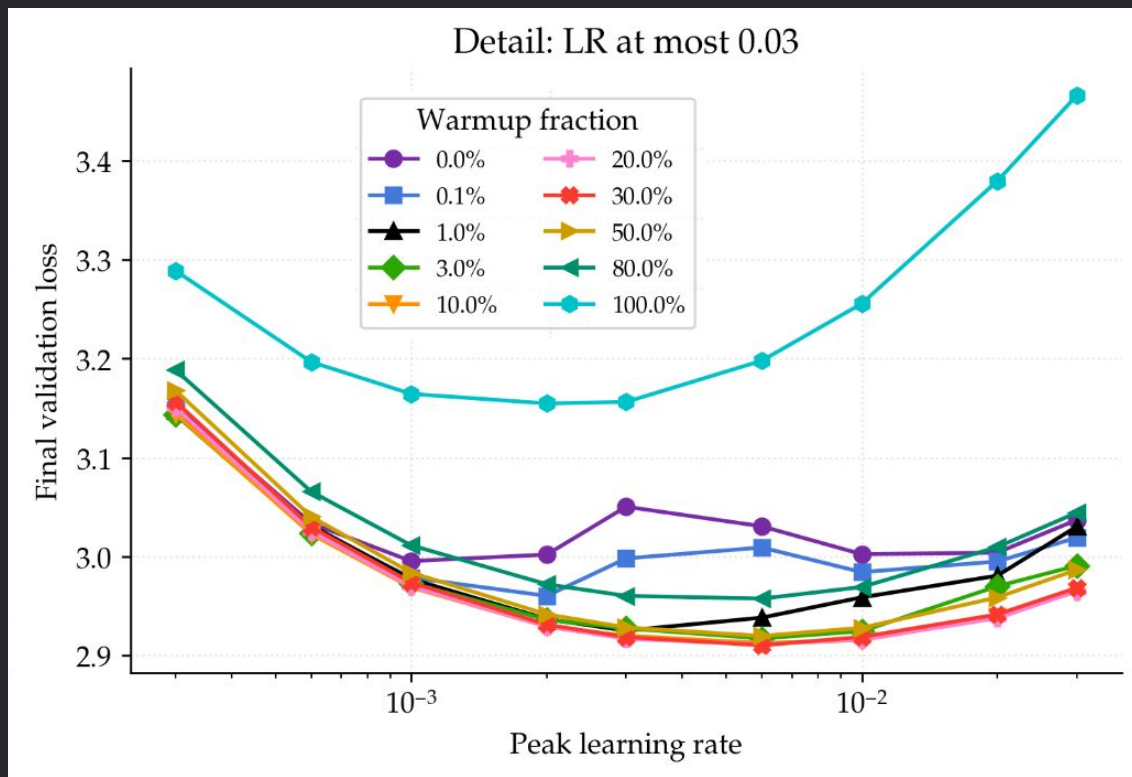


Question 5: Peak learning rate and warmup

Zoomed in on the range of interest.

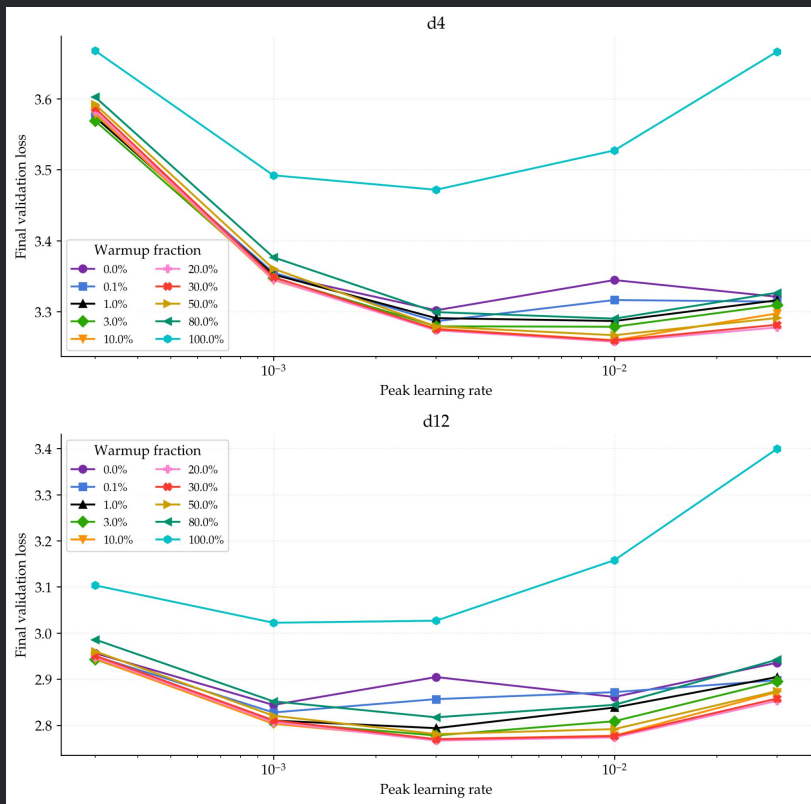
Warmup increases optimal LR and has smoother scaling

The optimal warmup for our setting is longer than 1%, closer to 20-30%. Maybe because our step count is so low?



Question 5: Peak learning rate and warmup

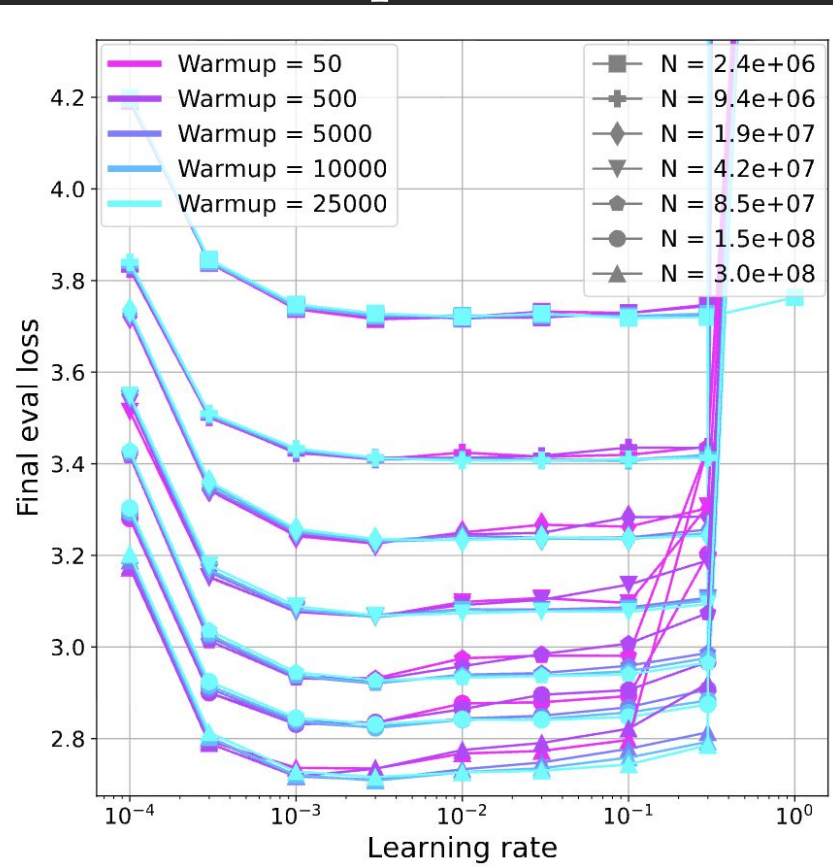
Seems to be similar situation for smaller/larger models on the same token horizon



Question 5: Peak learning rate and warmup

Previous work ([Wortsman et al., 2023](#)) has tried to characterize how various deep learning components improve stability (QKnorm, WD, warmup, muP, AdamW)

Their shapes have some similarities and differences



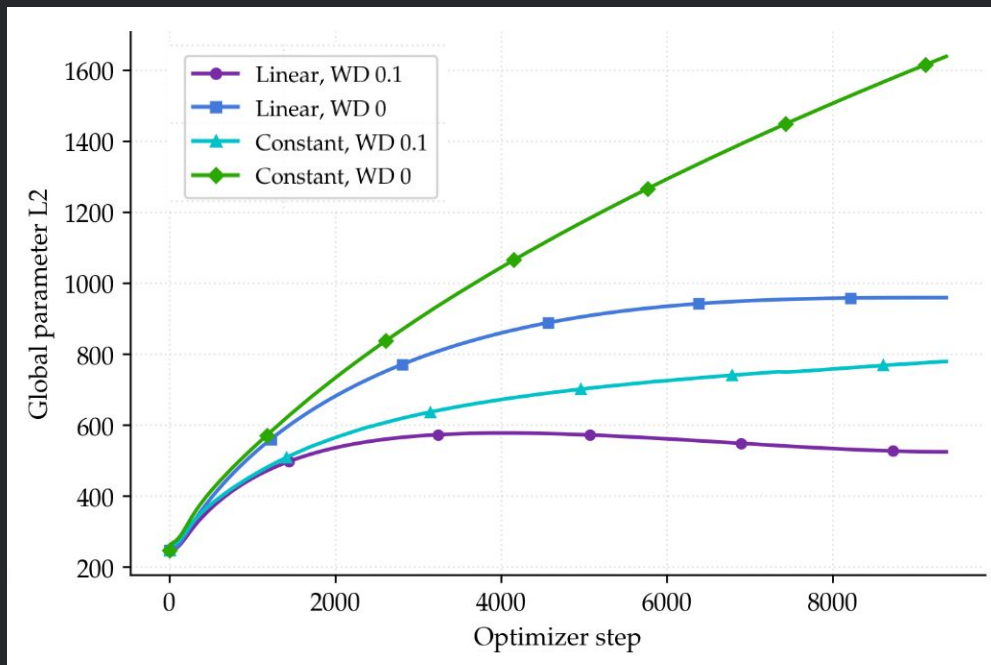
Question 5: Peak learning rate and warmup

Potential mental model updates

- Warmup improves tolerance to high learning rates and gives more stable scaling
- Model size does not play a huge role

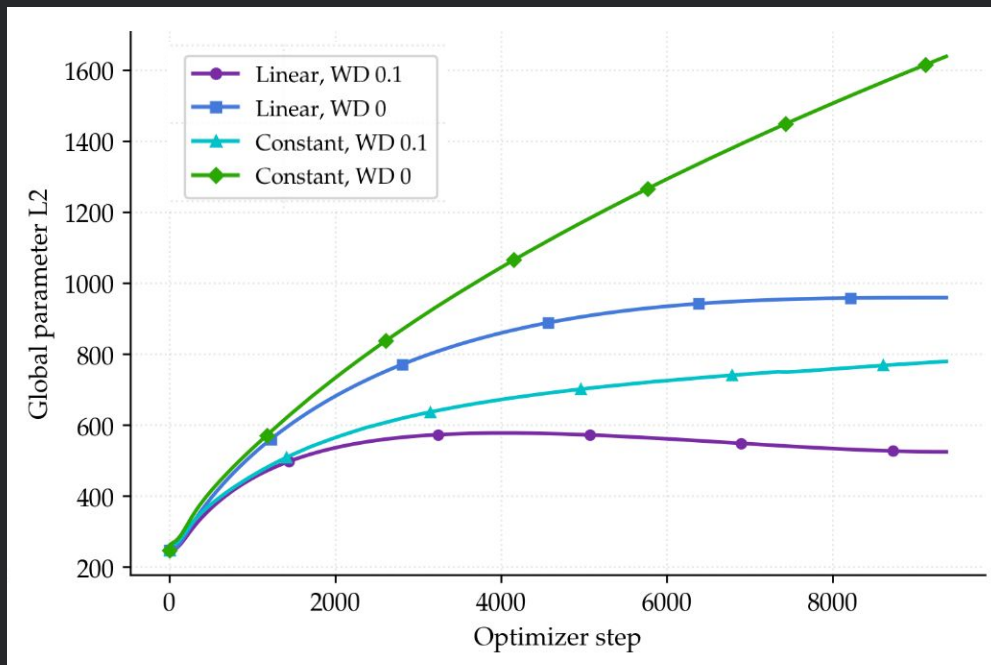
Question 6: Norm trajectories under schedule and WD

Answer 6a) Parameter norm is 3-15% below peak for linear 0.1 WD. Parameter norm is within 3% of peak for linear + 0.0 WD, constant + 0.0 WD, constant + 0.1 WD



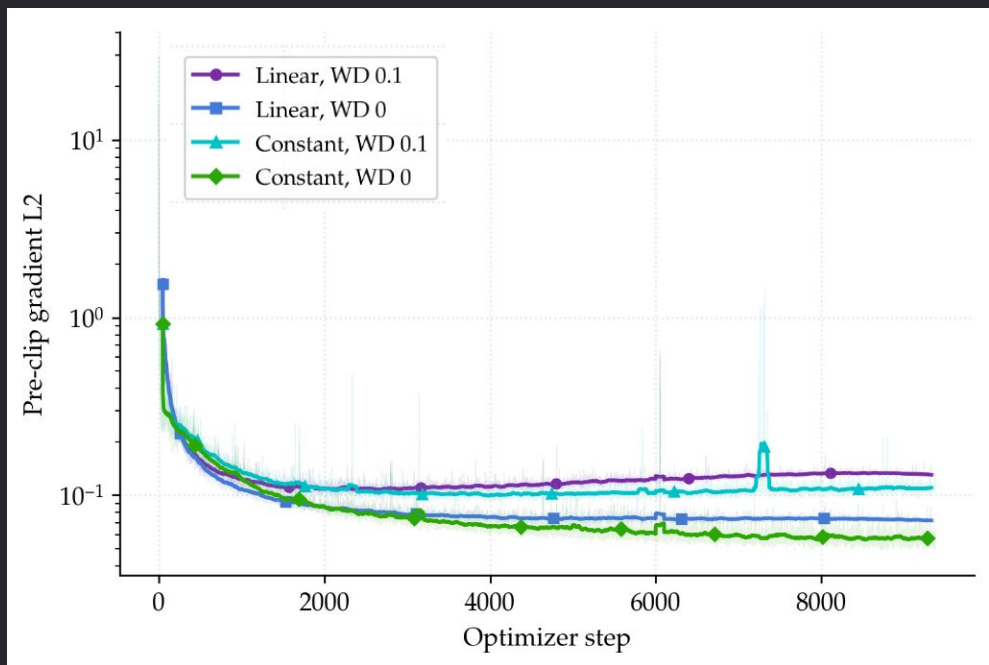
Question 6: Norm trajectories under schedule and WD

Lower lr seems to mean less parameter growth. Weight decay has same effect. Combination of both leads to parameter norm not growing constantly?



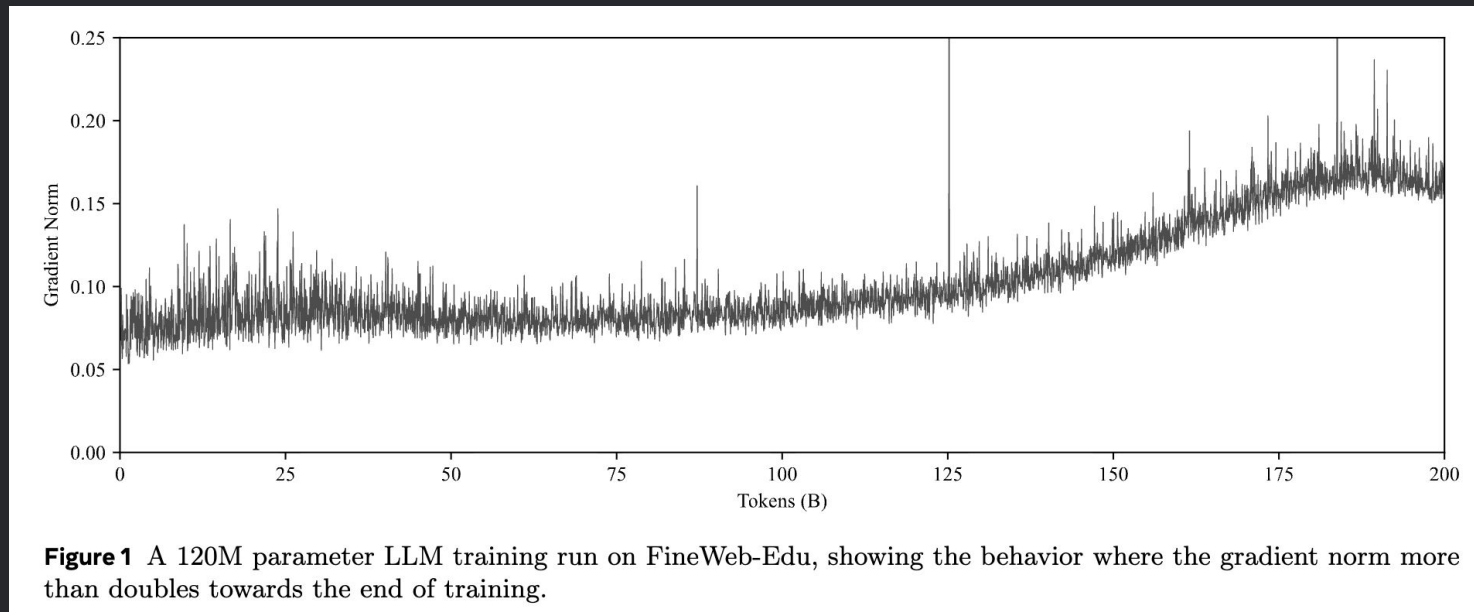
Question 6: Norm trajectories under schedule and WD

Answer 6b) None of the gradient norms double over the course of training



Question 6: Norm trajectories under schedule and WD

We do not see the phenomenon reported by AdamC ([Defazio 2025](#)) where gradient norm rapidly increases during training

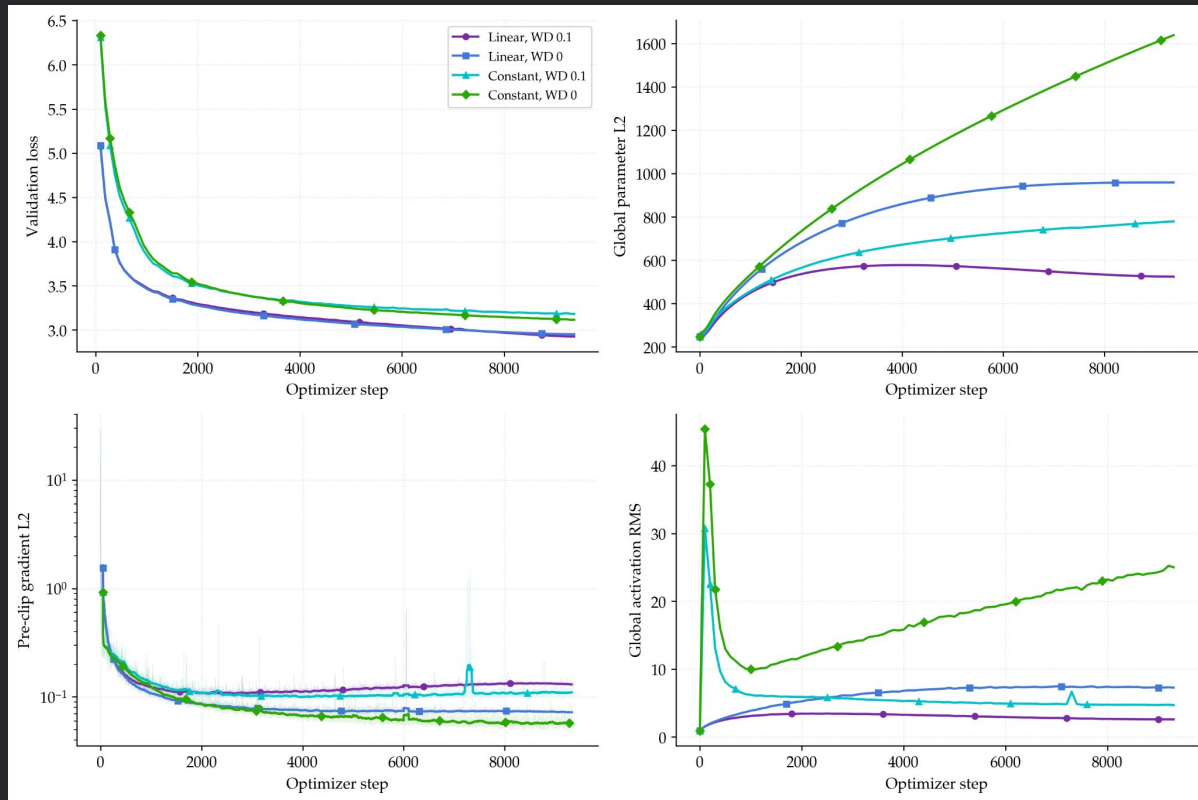


Question 6: Norm trajectories under schedule and WD

Loss and other parameter statistics

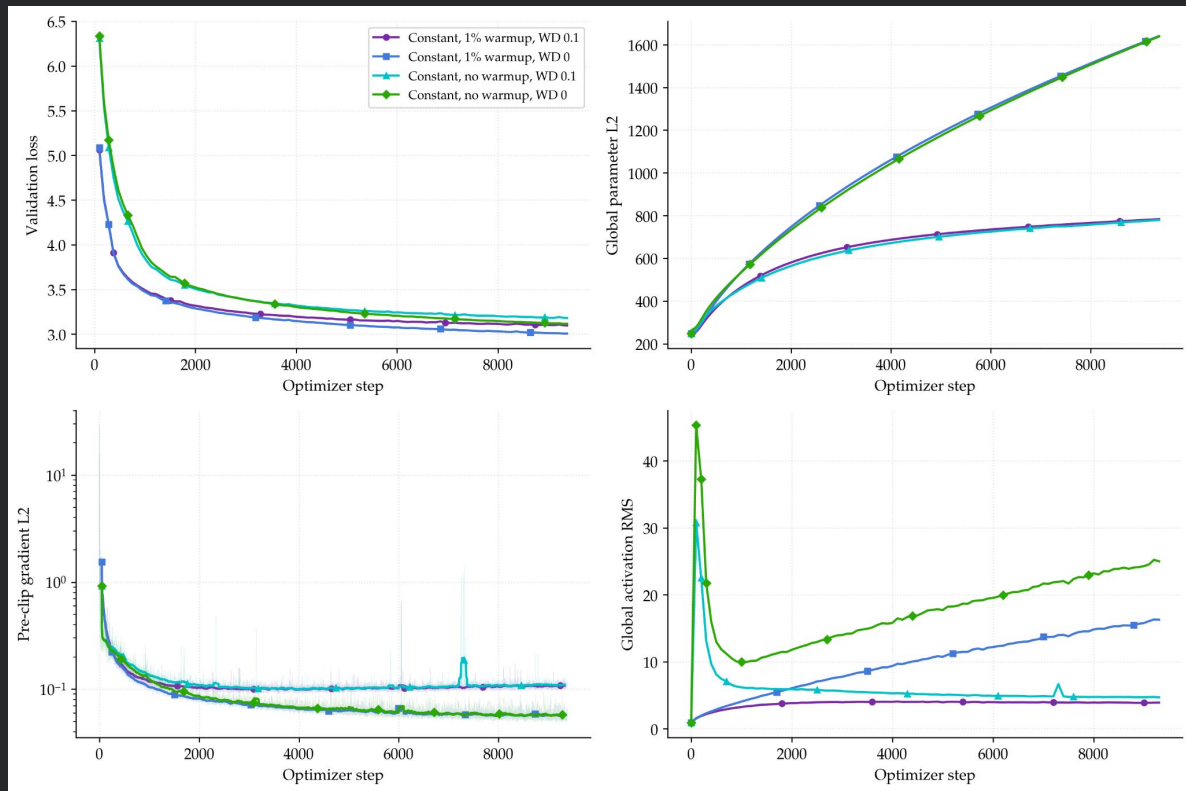
Gradient norm seems more driven by weight decay

Known effect that activation norm adversely affected by warmup



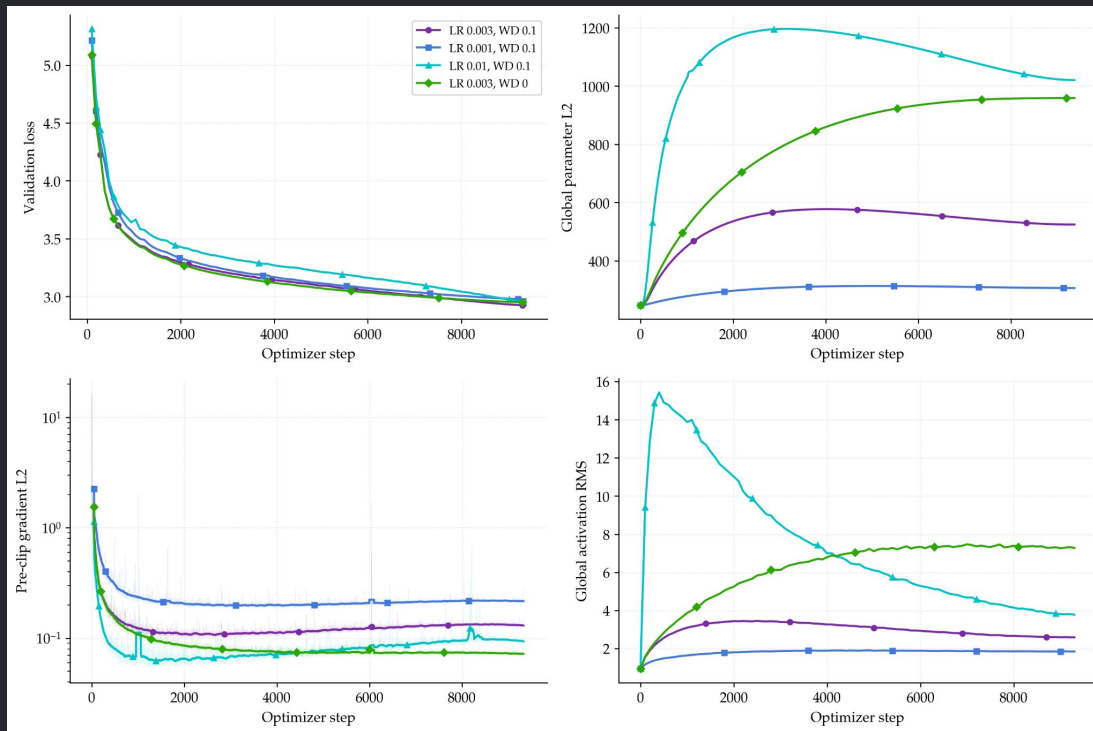
Question 6: Norm trajectories under schedule and WD

Constant's parameter norm is not affected by warmup even though its gradient norm is



Question 6: Norm trajectories under schedule and WD

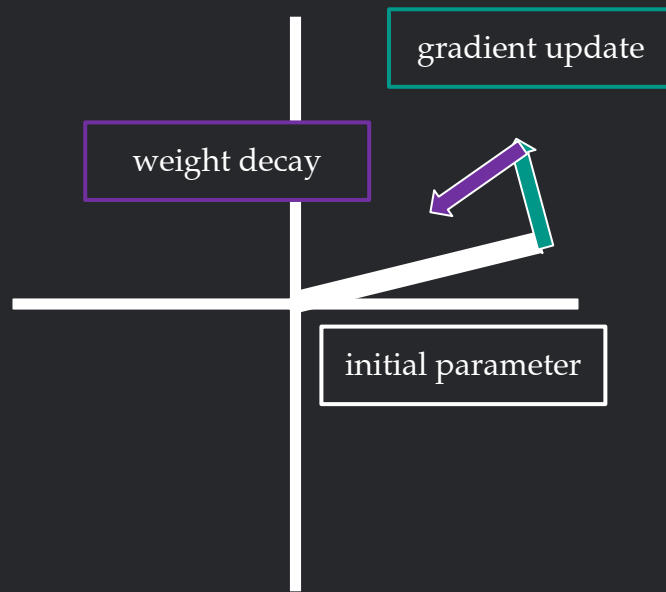
Different learning rates



Question 6: Norm trajectories under schedule and WD

[van Laarhoven 2017](#) argues that parameter norm should equilibrate over the course of training.

- Because of prenorm, almost all layers have the property $f(cw) = f(w)$
- In this case, gradient is orthogonal to weights (any change in the direction of the weights doesn't affect the loss). Therefore, parameter norm grows (and gradient norm shrinks even further)
- Weight decay pulls the parameter norm back to 0



Question 6: Norm trajectories under schedule and WD

Potential mental model updates

- Lower LR and higher weight decay control parameter norm, enough to even decrease it over course of training
- Warmup mostly affects activation norm and slightly affects gradient norm
- For constant LR, warmup may only affect loss and activation norm

Further reading: [van Laarhoven 2017](#)

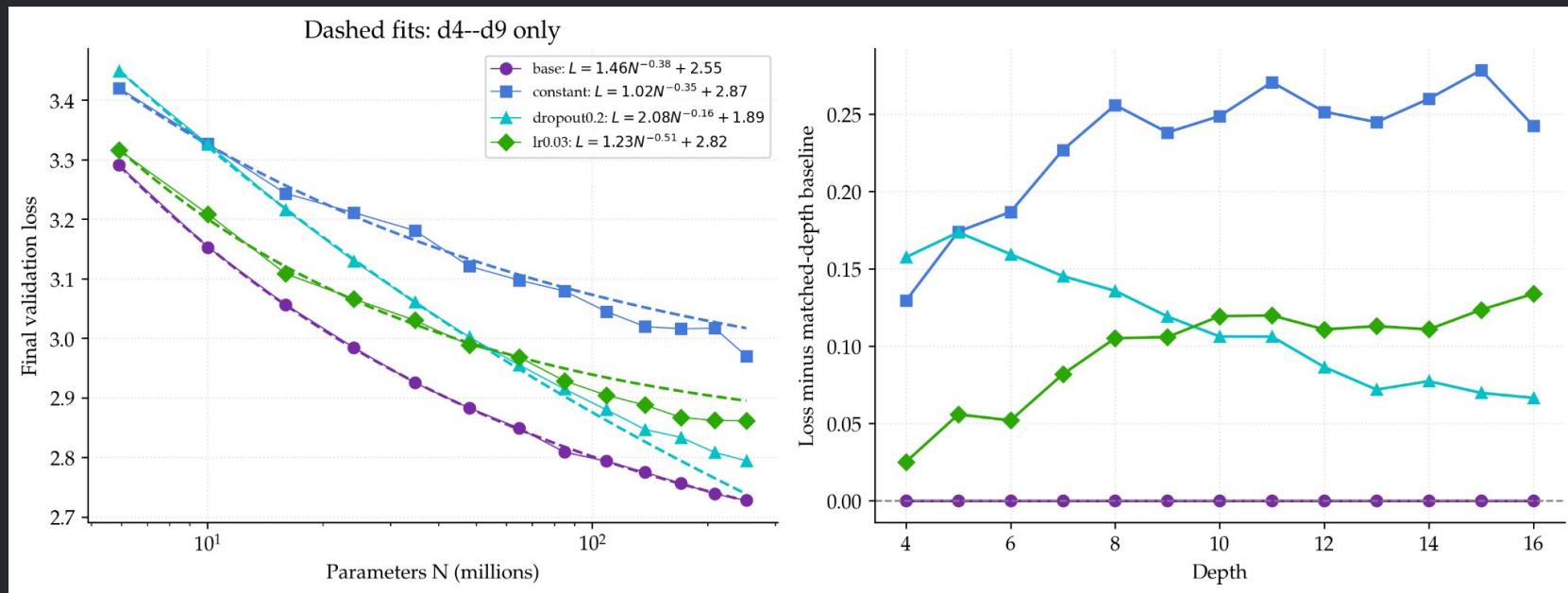
Question 7: Extrapolating your scaling ladders

Answer 7a) Dropout is fine, fixed high LR is okay, constant LR schedule is much worse

Recipe	d16 loss	Gap	Answer bin
Constant LR	2.97011	+0.24228	0.18–0.4
Dropout 0.2	2.79431	+0.06648	0–0.08
LR 0.03	2.86156	+0.13373	0.08–0.18

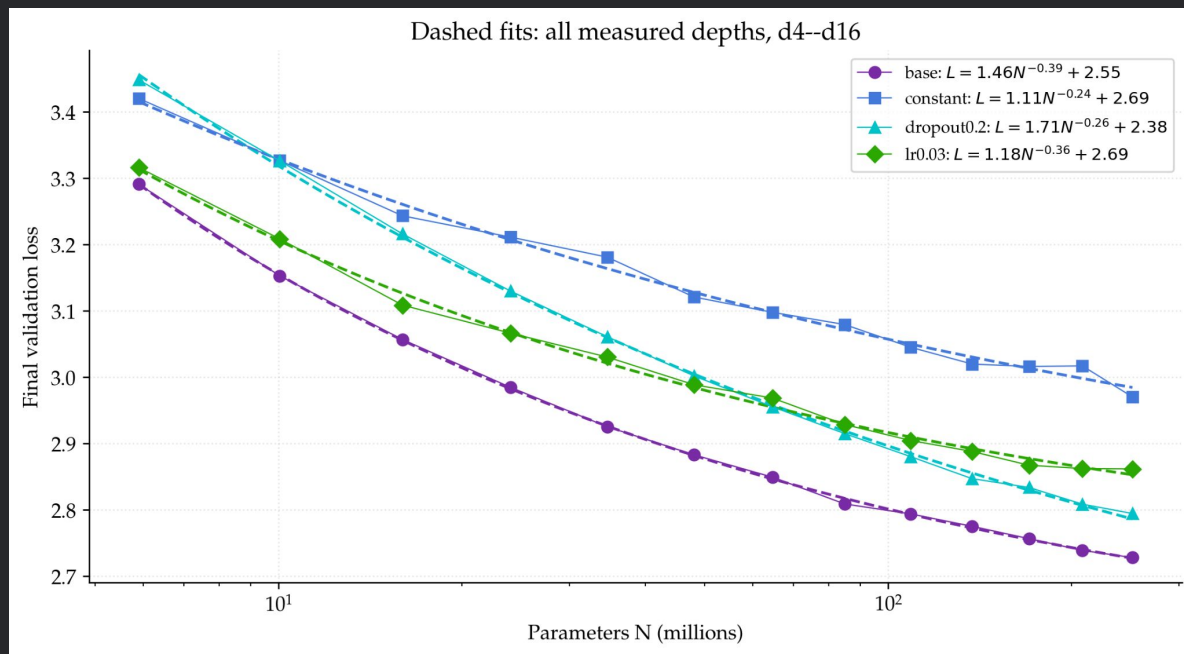
Question 7: Extrapolating your scaling ladders

Here are the scaling ladders for the assignment with fits through d4-d9 only. In this case, the fits are not perfect but do predict the rankings



Question 7: Extrapolating your scaling ladders

Meanwhile, the fit looks great if you use all the points in the ladder. Scaling laws are hard!



Question 7: Extrapolating your scaling ladders

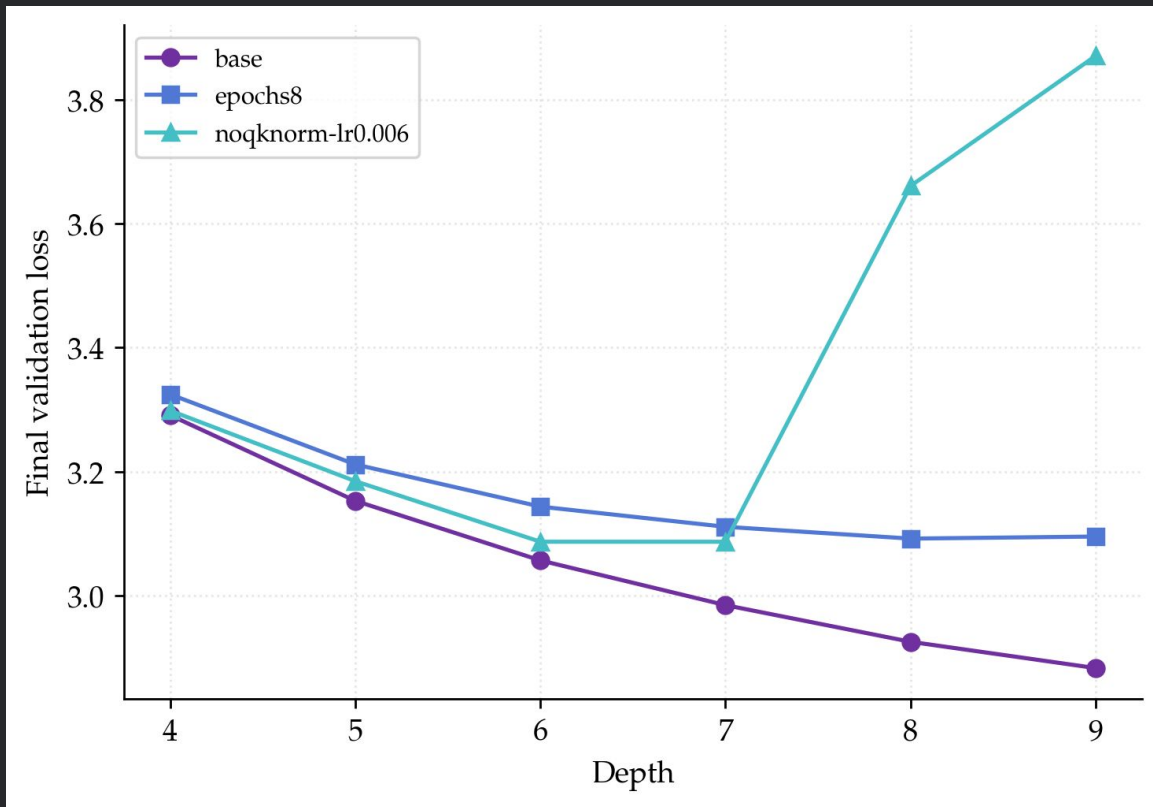
Answer 7b) d9 still helps for eight epochs but less than the reference 0.41. d9 significantly hurts for no QKNorm

Recipe	d4	d9	Difference	Answer
Baseline	3.29085	2.88283	-0.40802	Reference
Eight epochs	3.32460	3.09510	-0.22950	-0.25 to 0
No QK norm, LR 0.006	3.29834	3.87079	+0.57245	Above +0.5

Question 7: Extrapolating your scaling ladders

Larger models (relative to token count) eventually tolerate less epochs ([Kim et al, 2025](#))

QKNorm improves stability at high learning rates, important for larger models



Question 7: Extrapolating your scaling ladders

Potential mental model updates

- Different choices of hyperparameter recipes lead to different scaling, some cleaner than others
- Hard to ascertain quality of fit without extrapolation

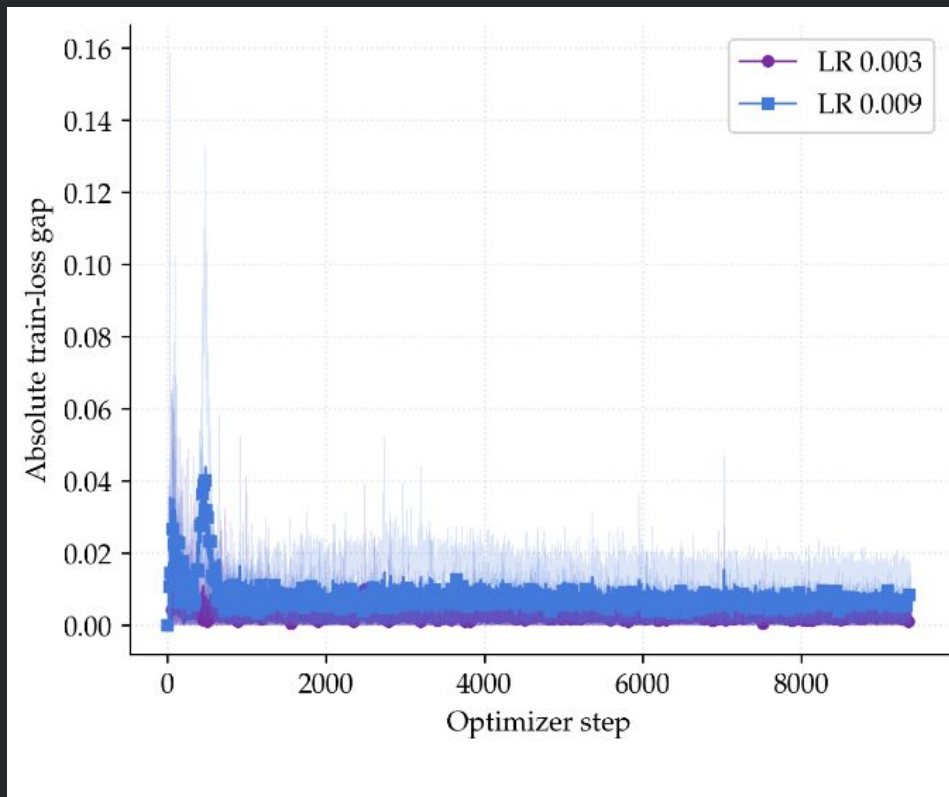
Question 8: One flipped token

Answer 8a) Five seed average of peak delta train loss was ~ 0.09 (0.05-0.16)

Seed	Peak train gap, LR .003	Final val gap, LR .003	Peak train gap, LR .009	Final val gap, LR .009
42	0.06251	0.002311	0.10270	0.005198
43	0.04615	0.000101	0.05200	0.006462
44	0.06397	0.000843	0.15840	0.013318
45	0.05937	0.001595	0.06940	0.005411
46	0.04115	0.001298	0.05838	0.002864
Mean	0.05463	0.001230	0.08817	0.006651

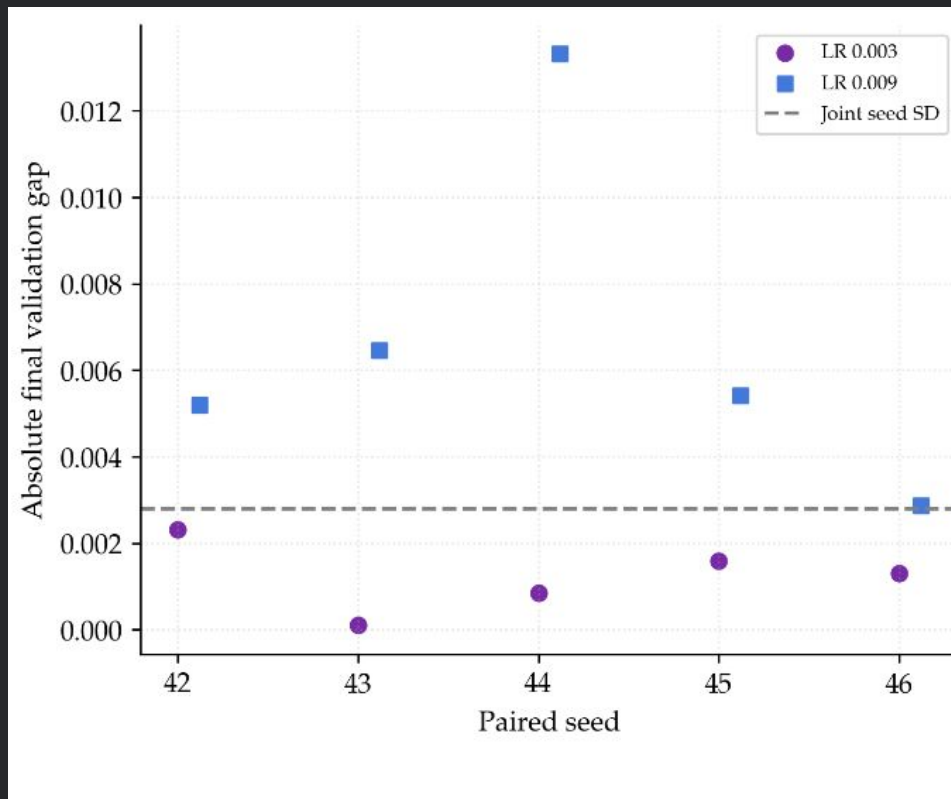
Question 8: One flipped token

Seems to happen early and then go away



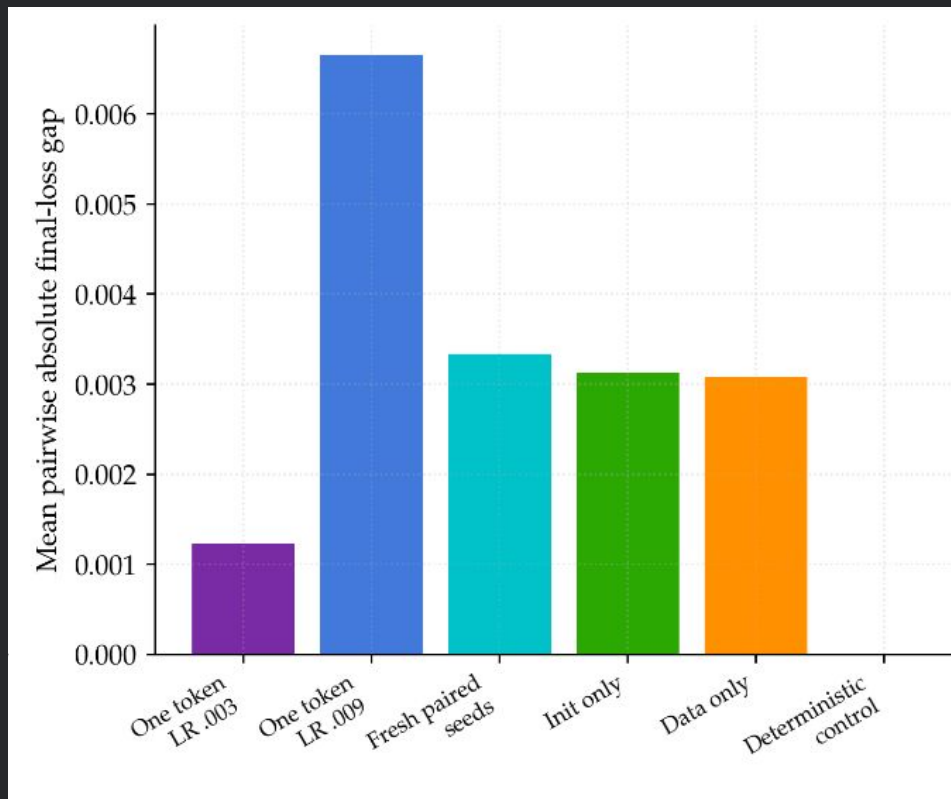
Question 8: One flipped token

Answer 8b) The standard deviation of final loss for lr 0.003 is within 0.2sigma and 5sigma (for standard loss standard deviation)



Question 8: One flipped token

Seems much higher for higher learning rate (note this is validation gap, not train gap)



Question 8: One flipped token

[Summers and Dineen 2021](#) show that flipping one initial parameter bit initial parameter basically gets all of the variance!

This agrees qualitatively but disagrees quantitatively (for matched learning rate, we got 3x lower SD)

Nondeterminism Source	PPL SD	Pairwise Disagree (%)	Ensemble PPL Δ
Parameter Initialization	0.20 ± 0.01	17.3	-2.07
Stochastic Operations	0.19 ± 0.01	17.3	-2.08
All Nondeterminism Sources	0.18 ± 0.01	17.4	-2.07

Nondeterminism Source	PPL SD	Pairwise Disagree (%)	Ensemble PPL Δ
Random Bit Change	0.19 ± 0.01	17.7	-2.07

Sneak peak for next assignment

- Understanding the relationship between learning rate and token horizon
- Understanding the relationship between learning rate and weight decay
- Testing the Noisy Quadratic Model for predictions on learning rate, batch size, and weight decay
- Testing the Maximal Update Parameterization for predictions on weight initialization and learning rate as a function of width/depth

Released by tonight. Will come with some instructors sweeps to save compute. While we're waiting for more concurrency, launch your runs early! Work with others (please please please), many positive examples so far

If you're on the waitlistwaitlist, email us for compute access